

Saad Ahmad

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PROFILE

AI researcher specializing in medical image analysis and deep learning for pelvic MRI, with experience across the full pipeline from raw DICOM acquisition to deployed model inference. Skilled in 3D U-Net and transformer-based architectures for anatomical landmark detection, segmentation, and automated clinical reporting workflows.

EDUCATION

Friedrich-Alexander-Universität Erlangen-Nürnberg

M.Sc. Data Science

Thesis: Automatic Detection of Anatomical Landmarks of the Sacral Spine and Uterus in MRI

Erlangen, Germany

09/2021 – 3/2026

International Islamic University

B.Sc. Software Engineering

Islamabad, Pakistan

10/2015 – 04/2019

PUBLICATIONS

Bhatia D., **Ahmad S.**, Tripathy S., Bustos Vivas M.C., Kratzsch L., Knupfer A., Aviles Verdera J., Hutter J. "Real-Time Automated Analysis and Reporting of Uterine Volumetry, Biometry, and Incidental Findings in Female Pelvic MRI." (Submitted, *IEEE Transactions on Medical Imaging*, 2025)

WORK EXPERIENCE

Smart Imaging Lab, Universitätsklinikum Erlangen

Research Assistant (Master's Thesis)

Live demo & deployments: [saadahmad.de](#)

Erlangen, Germany

09/2025 – 03/2026

- Designed a 3D dual-decoder U-Net for automatic lumbosacral vertebral landmark localization in pelvic MRI, achieving **4.21 mm mean localization error** across multi-center datasets.
- Developed dual- and triple-decoder 3D U-Net architectures for uterine landmark detection, achieving **4.52 mm mean landmark error** across three MRI acquisition protocols.
- Validated deep learning models on multi-center, multi-vendor pelvic MRI datasets (0.55T–3T) across heterogeneous clinical imaging protocols.

Softisans

AI Engineer

Remote

04/2024 – 02/2025

- Developed AI-powered applications using LLMs, OpenAI APIs, and React.js for intelligent user experiences.
- Built and integrated RESTful APIs, RAG pipelines, and semantic search systems for AI-driven solutions.

SKILLS

Deep Learning & AI: PyTorch · MONAI · nnUNet · Swin-UNETR · U-Net · Heatmap Regression · Medical Image Segmentation · DICOM · NIfTI

Programming & Deployment: Python · FastAPI · Docker · Linux · REST APIs · Git

AI Systems & APIs: OpenAI API · RAG Pipelines · Semantic Search · LLM Applications